PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Efficacy of education on injection technique for patients diagnosed with diabetes with lipohypertrophy: a systematic review and meta-analysis
AUTHORS	Ichikawa, Masahiro; Akiyama, Tomoaki; Tsujimoto, Yasushi; Anan, Keisuke; Yamakawa, Tadashi; Terauchi, Yasuo

VERSION 1 – REVIEW

REVIEWER	Sandro Gentile University of Campania Luigi Vanvitelli School of Medicine and Surgery, Internal Medicine
REVIEW RETURNED	10-Aug-2021

GENERAL COMMENTS	The topic of LH is important and still too widespread after about 100 years from the first reports in the literature. The paper is well written and the conclusions are quite in agreement with the literature data. However, as the authors used very broad criteria to select each type of manuscript, I think they still overlooked some interesting papers, listed below: 1. Sandro Gentile, Giuseppina Guarino, Teresa Della Corte, Giampiero Marino, Ersilia Satta, Maria Pasquarella, Carmine Romano, Carmelo Alfrone, Felice Strollo, AMD-OSDI Study Group on Injection Technique, Nefrocenter Research and Nyx Start-Up. Role of Structured Education in Reducing Lypodistrophy and its Metabolic Complications in Insulin-Treated People with Type 2 Diabetes: A Randomized Multicenter Case–Control Study Diabetes Ther. 2021 May; 12(5): 1379–1398. Published online 2021 Mar 18. doi: 10.1007/s13300-021-01006-0 2. Gentile S, Strollo F; Nefrocenter Research Study Group. Cost saving effects of a short-term educational intervention entailing lower hypoglycaemic event rates in people with type 1 diabetes and lipo-hypertrophy. Diabetes Res Clin Pract. 2018 Sep;143:320-321. doi: 10.1016/j.diabres.2018.07.030. Epub 2018 Aug 1. PMID: 30076872. 3. Matsumura M, Monden Y, Nakatani T. Improvement of glycemic control by re-education in insulin injection technique. Diabetes. 2007;56(S1):0157- OR.

	4. Smith M, Clapham L, Strauss K. UK Lipohypertrophy Interventional Study. Diabetes Res Clin Pract. 2017;126:248–53. https://doi.org/10.1016/j.diabres. 2017.01.020 (Epub 2017 Mar 2 PMID: 28288434). The second observation concerns the simultaneous inclusion of type 1 and type 2 patients, although several papers report important differences in frequency between the two types of diabetes. Additional comment: the text is long and especially the introduction must be simplified by avoiding mentioning well-known data in the literature, in order to make the text more impressive.

REVIEWER	Shahjada Selim Bangabandhu Sheikh Mujib Medical University, Endocrinology
REVIEW RETURNED	28-Sep-2021

GENERAL COMMENTS	This is a well written systematic review.

REVIEWER	Julia Blanchette Cleveland Clinic, Endocrinology and Metabolism, Diabetes Center
REVIEW RETURNED	30-Sep-2021

GENERAL COMMENTS	Abstract -The design section is not a complete sentence. General -This is a novel and relevant literature review on a topic that is often seen clinically in diabetes care and education, but not discussed enough in research. The review design is well organized and the analysis is thorough. I applaud the authors for writing this paper in English, and most of it is written clearly and excellent. However, there are a couple of minor grammatical errors throughout so I recommend doing another round of edits.

REVIEWER	Felix Strollo San Raffaele Research Institute
REVIEW RETURNED	05-Nov-2021

GENERAL COMMENTS	It is a real pity that only two papers met the requirements for the meta-analysis, so that no really sound conclusions could be drawn from such a big effort. However, a severe limitation comes from the fact that the two studies involve both types of diabetes..... It is widely accepted that the two types cannot be mixed up in studies concerning insulin-induced lipodystrophies, due to the large difference in the daily injection number between them and other intrinsic differences so that only one type should be analyzed at a time. Also, another RCT has been published quite recently (Gentile S et al. Role of Structured Education in Reducing Lypodistrophy and its Metabolic Complications in Insulin-Treated People with Type 2 Diabetes: A Randomized Multicenter Case–Control Study, Diabetes Ther (2021) 12:1379–1398) which was not included and might help increase the number of patients after limiting them to T2DM. It might well be that, after pooling data from T2DM patients from the three

	papers, the Authors can draw sounder conclusions .....

VERSION 1 – AUTHOR RESPONSE

Reviewer #1:

Comment 1. The topic of LH is important and still too widespread after about 100 years from the first reports in the literature.
The paper is well written and the conclusions are quite in agreement with the literature data.
However, as the authors used very broad criteria to select each type of manuscript, I think they still overlooked some interesting papers, listed below:

1. Sandro Gentile, Giuseppina Guarino, Teresa Della Corte, Giampiero Marino, Ersilia Satta, Maria Pasquarella, Carmine Romano, Carmelo Alfrone, Felice Strollo, AMD-OSDI Study Group on Injection Technique, Nefrocenter Research and Nyx Start-Up. Role of Structured Education in Reducing Lypodistrophy and its Metabolic Complications in Insulin-Treated People with Type 2 Diabetes: A Randomized Multicenter Case–Control Study
Diabetes Ther. 2021 May; 12(5): 1379–1398. Published online 2021 Mar 18. doi: 10.1007/s13300-021-01006-0
2. Gentile S, Strollo F; Nefrocenter Research Study Group. Cost saving effects of a short-term educational intervention entailing lower hypoglycaemic event rates in people with type 1 diabetes and lipo-hypertrophy. Diabetes Res Clin Pract. 2018 Sep;143:320-321. doi: 10.1016/j.diabres.2018.07.030. Epub 2018 Aug 1. PMID: 30076872.
3. Matsumura M, Monden Y, Nakatani T. Improvement of glycemic control by re-education in insulin injection technique. Diabetes. 2007;56(S1):0157- OR.
4. Smith M, Clapham L, Strauss K. UK Lipohypertrophy Interventional Study. Diabetes Res Clin Pract. 2017;126:248–53. https://doi.org/10.1016/j.diabres. 2017.01.020 (Epub 2017 Mar 2 PMID: 28288434).

Response: We appreciate the reviewer's remark and comment. We missed reference 1 above because we searched before March 18, 2021, hence we have performed a search update on November 11, 2021. The eligibility criteria for our systematic review were limited to RCTs in diabetic patients with LH in all cases, so references 2 through 4 did not meet the eligibility criteria.

In accordance with the update, we have revised the entire manuscript.

Comment 2. The second observation concerns the simultaneous inclusion of type 1 and type 2 patients, although several papers report important differences in frequency between the two types of diabetes.

Response: We appreciate the reviewer's suggestion. In accordance with the comment, we tried to perform a post-hoc subgroup analysis of types of DM. However, we were unable to do this because the outcome data was not reported separately for the types of DM. As such, we have added it as a limitation of our study.

Change: We have added the above as the third limitation as follows. (page 22, para 2)

"Third, we could not perform a subgroup analysis of the types of DM which was requested during peer-review due to the limited data. The prevalence of LH was higher in type 2 diabetes than in type 1 diabetes, thus the types of DM may be an effect modifier of IT. [4] Future studies that use individual participant data can assess the effect modification.."

4 Deng N, Zhang X, Zhao F, et al. Prevalence of lipohypertrophy in insulin-treated diabetes patients: A systematic review and meta-analysis. *J Diabetes Investig* 2017;9(3):536–543.

Comments 3. the text is long and especially the introduction must be simplified by avoiding mentioning well-known data in the literature, in order to make the text more impressive.

Response: We appreciate the reviewer's valuable suggestion. Accordingly, we have simplified the introduction and results section.

Change: We have shortened the introduction (page 5-6) and shortened the study characteristics in the results section. (page 14-15)

Reviewer #2:

Comment 1. This is a well written systematic review.

Response: We thank the reviewer for the positive comment.

Reviewer #3:

Comment 1.

Abstract
-The design section is not a complete sentence.

General

-This is a novel and relevant literature review on a topic that is often seen clinically in diabetes care and education, but not discussed enough in research. The review design is well organized and the analysis is thorough. I applaud the authors for writing this paper in English, and most of it is written clearly and excellent. However, there are a couple of minor grammatical errors throughout so I recommend doing another round of edits..

Response: We thank the reviewer for the positive comment. We have requested professional editing services of the manuscript from Editage (www.editage.jp) English language editing service.

Reviewer #4:

Comment 1. It is a real pity that only two papers met the requirements for the meta-analysis, so that no really sound conclusions could be drawn from such a big effort. However, a severe limitation comes from the fact that the two studies involve both types of diabetes..... It is widely accepted that the two types cannot be mixed up in studies concerning insulin-induced lipodystrophies, due to the large difference in the daily injection number between them and other intrinsic differences so that only one type should be analyzed at a time. Also, another RCT has been published quite recently (Gentile S et al. Role of Structured Education in Reducing Lypodistrophy and its Metabolic Complications in Insulin-Treated People with Type 2 Diabetes: A Randomized Multicenter Case–Control Study, Diabetes Ther (2021) 12:1379–1398) which was not included and might help increase the number of patients after limiting them to T2DM. It might well be that, after pooling data from T2DM patients from the three papers, the Authors can draw sounder conclusions .....

Response: We appreciate the reviewer's suggestions. We have performed a search update and added the RCT (Gentile S et al. Role of Structured Education in Reducing Lypodistrophy and its Metabolic Complications in Insulin-Treated People with Type 2 Diabetes: A Randomized Multicenter Case–Control Study, Diabetes Ther (2021) 12:1379–1398). In accordance with the comment, we tried to perform a post-hoc subgroup analysis of the types of DM. However, we were unable to do this because the outcome data was not reported separately for the types of DM. Accordingly, we have added it as a limitation of our study.

Change: We have performed a search update adding the reference above. We evaluated

the risk of bias of the newly added reference and added it to the meta-analysis.

We have therefore revised the entire manuscript. We have added the above asthe third limitation as follows. (page 22, para 2)

“Third, we could not perform a subgroup analysis of types of DM which was requested during peer-review due to the limited data. The prevalence of LH was higher in type 2 diabetes than in type 1 diabetes, thus the types of DM may be an effect modifier of IT. [4] Future studies that use individual participant data can assess the effect modification.

VERSION 2 – REVIEW

REVIEWER	Sandro Gentile University of Campania Luigi Vanvitelli School of Medicine and Surgery, Internal Medicine
REVIEW RETURNED	08-Dec-2021

GENERAL COMMENTS	Can the authors consider a recent study on the role of education that extends the observation period to 12 months? "The Durability of an Intensive, Structured Education-Based Rehabilitation Protocol for Best Insulin Injection Practice: The ISTERP-2 Study. Diabetes Ther. 2021 Sep;12(9):2557-2569. doi: 10.1007/s13300-021-01108-9. Epub 2021 Aug 12. PMID: 34383261; PMCID: PMC8385007" Without altering the structure of the paper, the authors could post a comment on this research that provides longer-lasting information?

REVIEWER	Felix Strollo San Raffaele Research Institute
REVIEW RETURNED	14-Dec-2021

GENERAL COMMENTS	Although technically possible, a metaanalysis of only 3 nonhomogenous studies (which even ignores a recent 12-month duration RCT!) is scientifically unacceptable, especially considering that it deals with only one RCT and two smaller studies concerning both T1DM and T2DM, a wrong choice per se, being the two types completely different under many aspects, including lipohypertrophy risk. This, in the end, only allowed conclusions on a very short follow-up. Such a choice obviously led to very limited results, so that conclusions are misleading by ignoring many widely accepted clinical advantages of correct injection techniques only based on the choice of papers and the consequent limited number of patients. Moreover, when dealing with TDD reduction, the authors reverse reasoning: it is nonsense to discuss the adverse effects of insulin excess here without underlining the important positive consequence of using a correct injection technique, i.e., improved insulin pharmacodynamics, and pharmacokinetics requiring reduced insulin dosage per se... So I suggest the Authors to go deeper into the subject before engaging in such a difficult task, requiring great experience and clinical dedication, and potentially downplayed by conclusions drawn only on non-clinically-supported statistical grounds.

VERSION 2 – AUTHOR RESPONSE

Reviewer #1:

Comment 1. Can the authors consider a recent study on the role of education that extends the observation period to 12 months? "The Durability of an Intensive, Structured Education-Based Rehabilitation Protocol for Best Insulin Injection Practice: The ISTERP-2 Study. Diabetes Ther. 2021 Sep;12(9):2557-2569. doi: 10.1007/s13300-021-01108-9. Epub 2021 Aug 12. PMID: 34383261; PMCID: PMC8385007"
Without altering the structure of the paper, the authors could post a comment on this research that provides longer-lasting information?

Response: We thank you for this relevant paper. Please note that we have mentioned in the limitation part that repeated education may have a long-lasting effect.

Changes: We have added the following part to the revised manuscript:

'Fourth, we could not evaluate the effects beyond 6 months in this study. The two studies, i.e., ISTERP-1 [21] and ISTERP-2,[25] had a 12-month follow-up duration, and although the ISTERP-2 study did not meet the eligibility criteria of this study, their findings suggest that repeating the education on injection technique has lasting effects.[21, 25]'

Reviewer #4

Comment 1. Although technically possible, a metaanalysis of only 3 nonhomogenous studies (which even ignores a recent 12-month duration RCT!) is scientifically unacceptable, especially considering that it deals with only one RCT and two smaller studies concerning both T1DM and T2DM, a wrong choice per se, being the two types completely different under many aspects, including lipohypertrophy risk. This, in the end, only allowed conclusions on a very short follow-up. Such a choice obviously led to very limited results, so that conclusions are misleading by ignoring many widely accepted clinical advantages of correct injection techniques only based on the choice of papers and the consequent limited number of patients. Moreover, when dealing with TDD reduction, the authors reverse reasoning: it is nonsense to discuss the adverse effects of insulin excess here without underlining the important positive consequence of using a correct injection technique, i.e., improved insulin pharmacodynamics, and pharmacokinetics requiring reduced insulin dosage per se... So I suggest the Authors to go deeper into the subject before engaging in such a difficult task, requiring great experience and clinical dedication, and potentially downplayed by conclusions drawn only on non-clinically-supported statistical grounds.

Response: We appreciate your constructive criticism. Although a recent 12-month duration RCT did not meet the eligibility criteria and was thus not included in our-meta-analysis, we have stated that repeated education may have a long-lasting effect. Especially, we have added the following part to the revised manuscript:

'Fourth, we could not evaluate the effects beyond 6 months in this study. The two studies, i.e., ISTERP-1 [21] and ISTERP-2,[25] had a 12-month follow-up duration, and although the ISTERP-2 study did not meet the eligibility criteria of this study, their findings suggest that repeating the education on injection technique has lasting effects.[21, 25]'

We agree with your point that we should focus on clinical aspects rather than statistical results. To incorporate the clinical aspects and the uncertainty of the results into our meta-analysis, we used the GRADE approach. For example, we rated down the certainty of evidence because we observed high heterogeneity in the included studies, which might be attributed to the types of diabetes. Indeed, two of the three main outcomes were rated as very low certainty of evidence. This meant that the evidence is very uncertain concerning the effect of injection technique education on such outcomes. [1] Thus, our conclusions are based not only on the statistics but also on clinical aspects.

As per your comment, we have mentioned the important positive consequence of using a correct inje ction technique in the Discussion section as follows:

‘The present study showed a decrease in insulin TDD of 6.26 U/day with improved pharmacokinetics/pharmacodynamics by injection technique education. Therefore, although the effects obtained by exercise interventions, which have different mechanism, showed greater results, injection technique education may be effective in avoiding excessive doses of insulin.’